

Week 3 Day 1 Academic And Workforce Foundations

기준 연결

기준	Day 1 적용	학생 evidence
ABET problem analysis	서비스 의존성과 장애 원인 분류	topology/failure table
ABET communication	개발팀과 운영팀이 공유할 장애 리포트 작성	handoff note
CS2023 Knowledge	distributed systems, networking, reliability 개념	service map
CS2023 Skill	Compose 실행, 로그 확인, 설정 변경	command evidence
NIST NICE	credential hygiene, least privilege, operational monitoring	config/security note
SRE practice	cascading failure, postmortem, health check	RCA note

핵심 판단

MSA 학습의 핵심은 서비스를 많이 쪼개는 것이 아니라, 쪼갬 서비스의 실행 조건과 장애 영향을 설명할 수 있는가이다. 배포 단위가 늘면 팀 독립성과 확장성의 가능성이 생기지만, 네트워크 실패와 로그 분산, 설정 관리 비용도 늘어난다.

Official References

Topic	Reference	확인할 키워드
Microservices on AWS	https://docs.aws.amazon.com/whitepapers/latest/microservices-on-aws/microservices.html	service, API, database, deployment
AWS Microservices	https://aws.amazon.com/microservices/	independent component, business capability
Martin Fowler Microservices Guide	https://martinfowler.com/microservices/	independently deployable, lightweight communication
Docker Compose	https://docs.docker.com/compose/	services, networks, depends_on, healthcheck
Compose services reference	https://docs.docker.com/reference/compose-file/services/	healthcheck, depends_on, environment
Google SRE Cascading Failures	https://sre.google/sre-book/addressing-cascading-failures/	failure propagation, overload, mitigation
OpenTelemetry Concepts	https://opentelemetry.io/docs/concepts/	traces, metrics, logs, observability
Twelve-Factor App	https://12factor.net/	config, backing services, logs